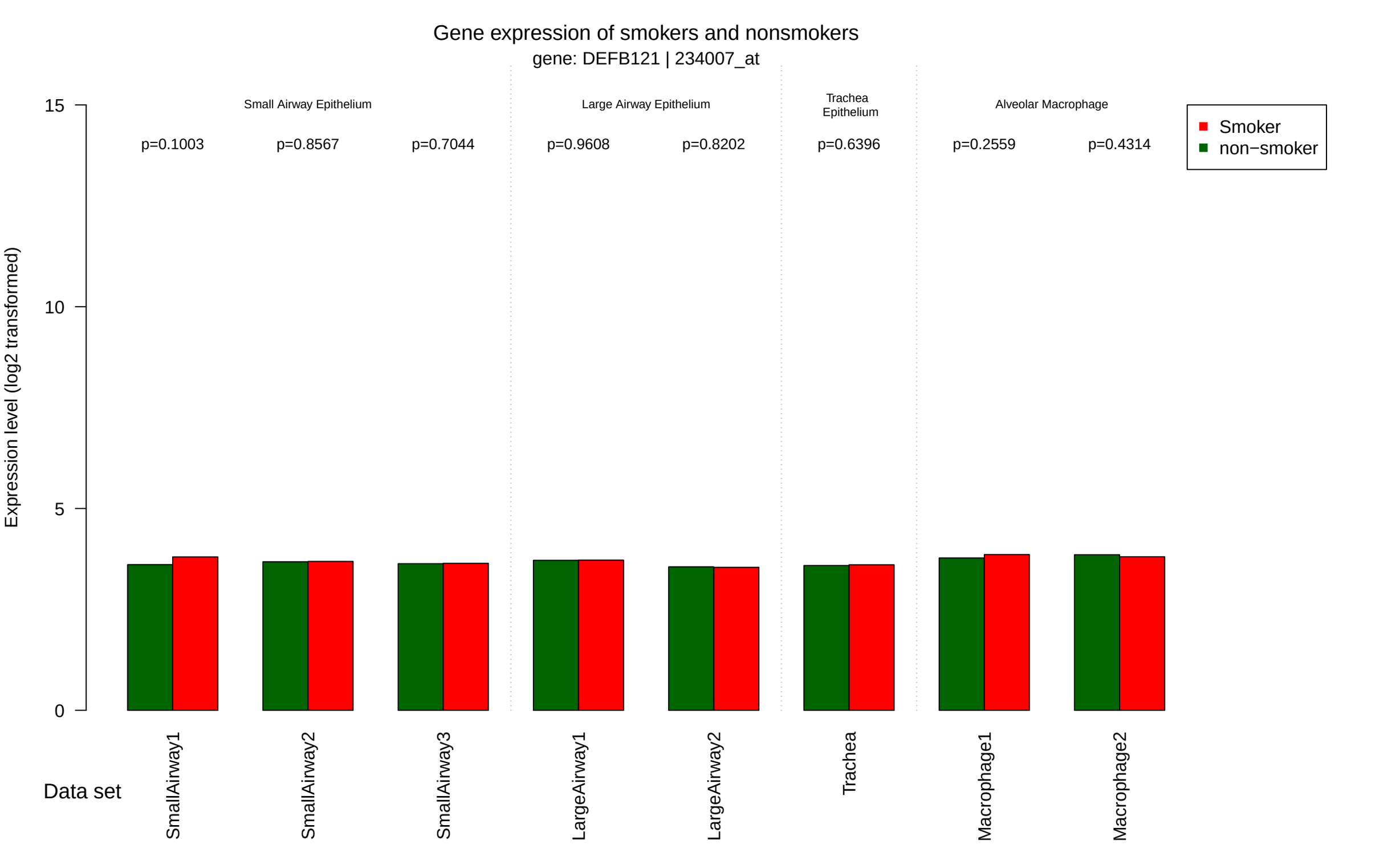
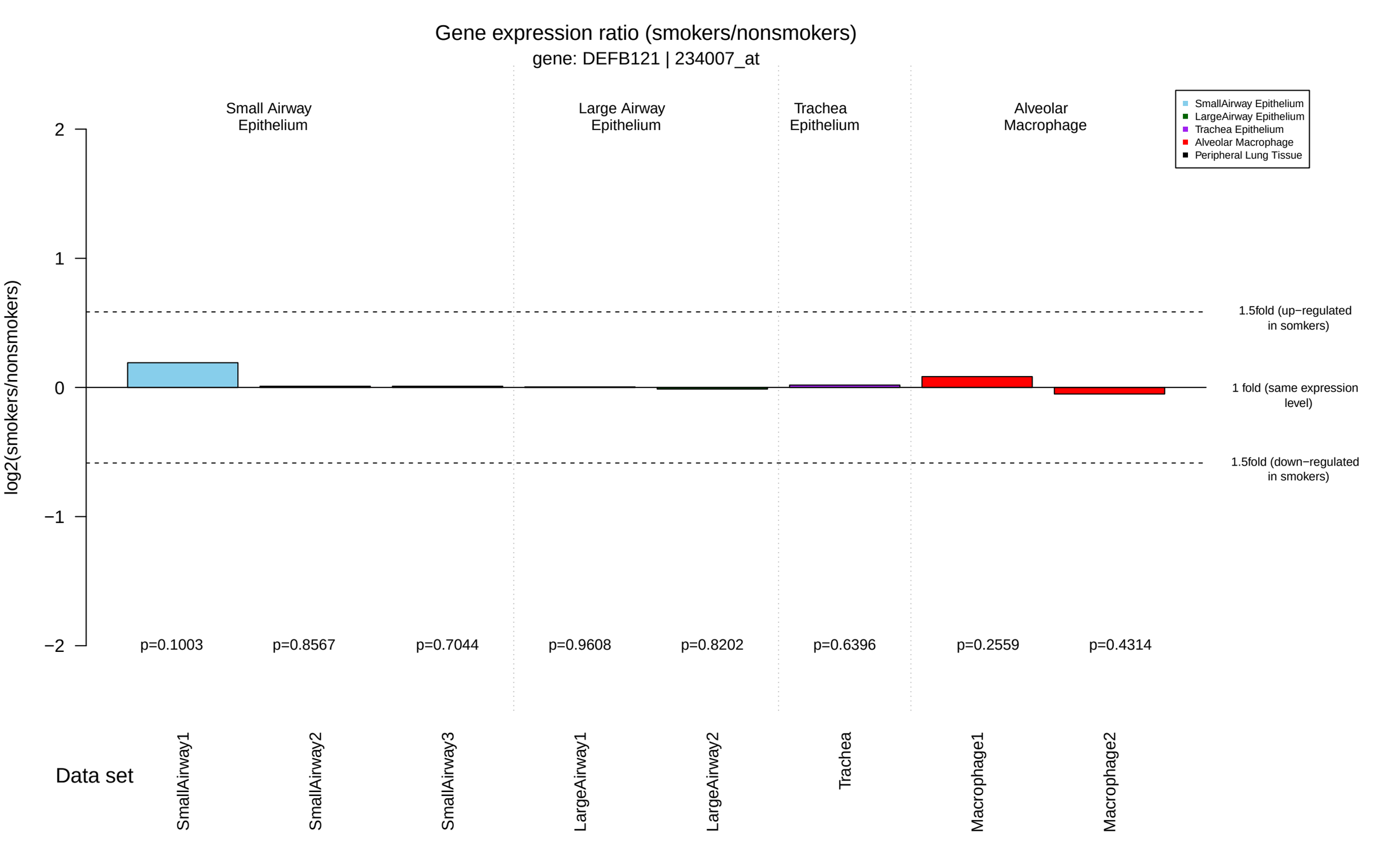
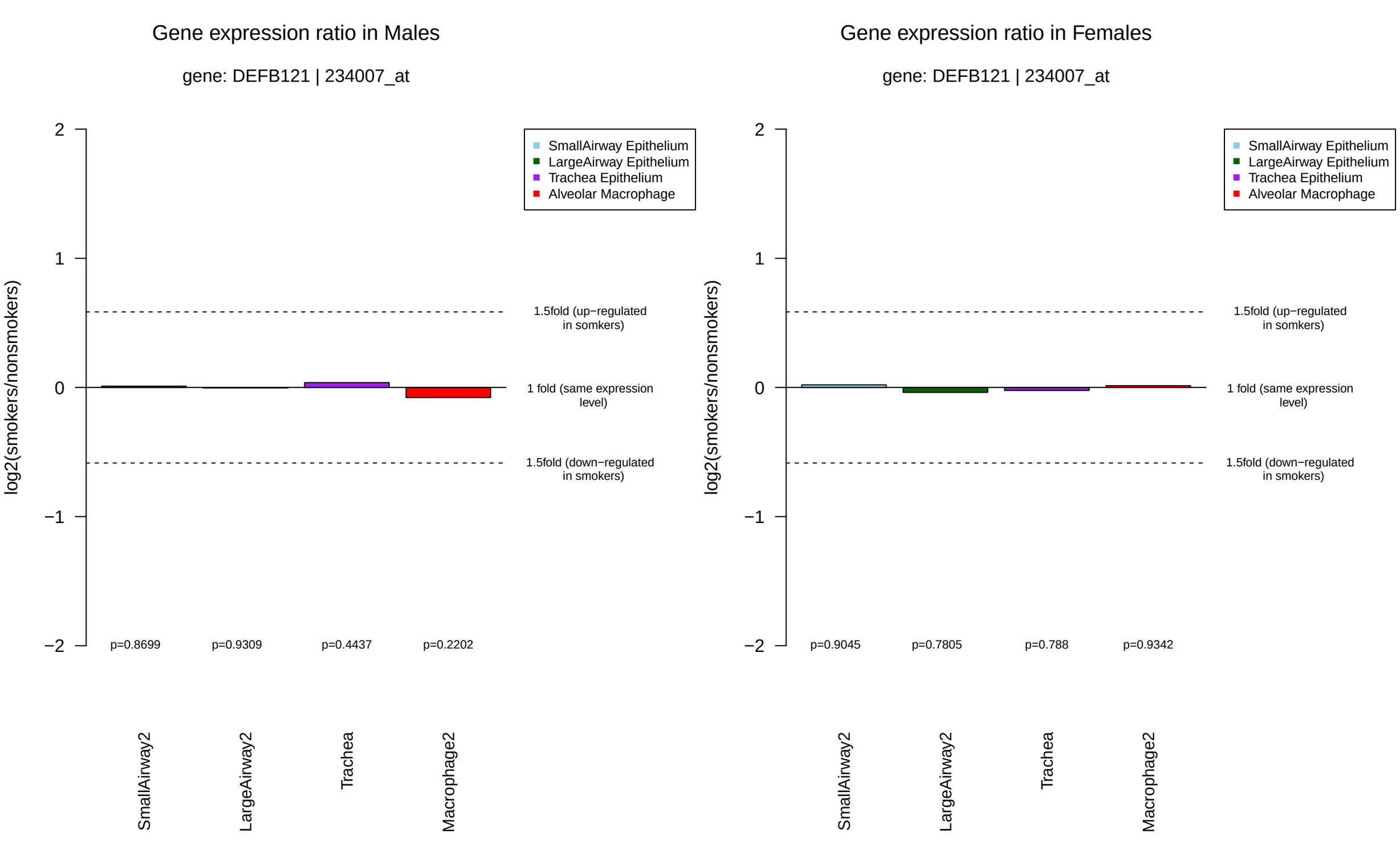
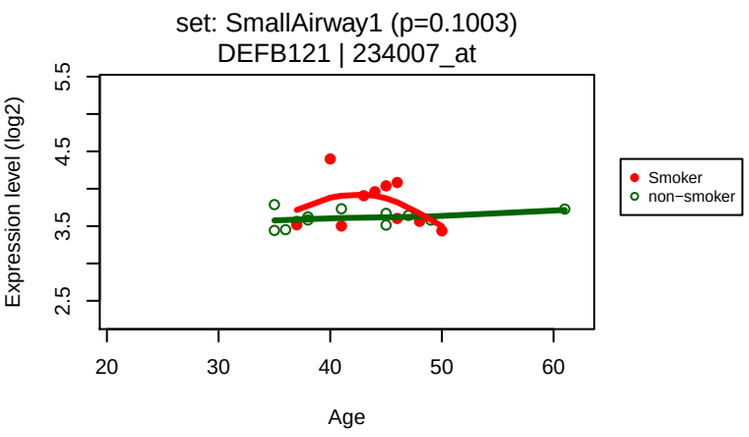
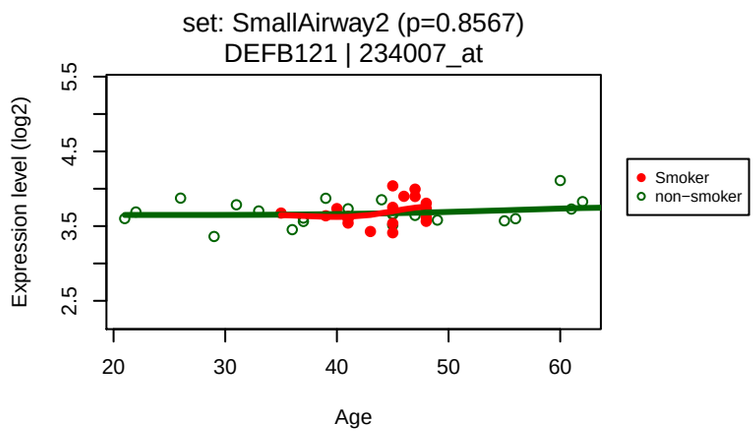
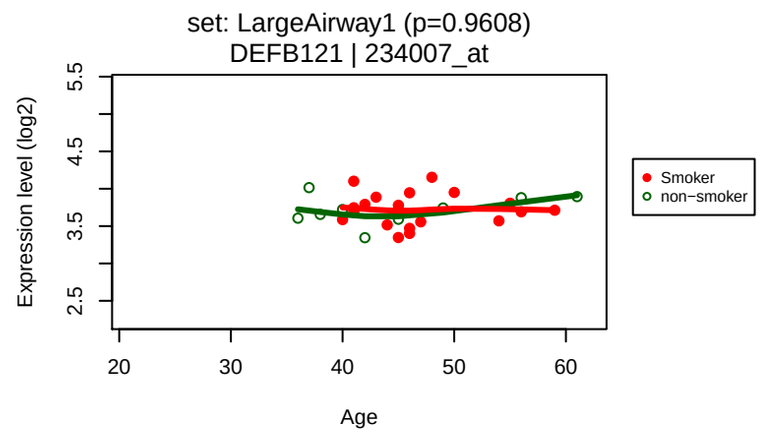

Gene expression of smokers and nonsmokers
gene: DEFB121 | 234007_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1003
p=0.8567
p=0.7044
p=0.9608
p=0.8202
p=0.6396
p=0.2559
p=0.4314
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

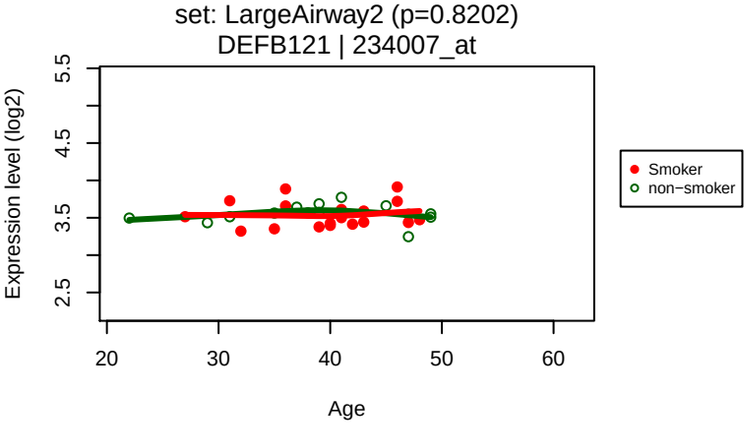

Gene expression ratio (smokers/nonsmokers)
gene: DEFB121 | 234007_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1003
p=0.8567
p=0.7044
p=0.9608
p=0.8202
p=0.6396
p=0.2559
p=0.4314
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

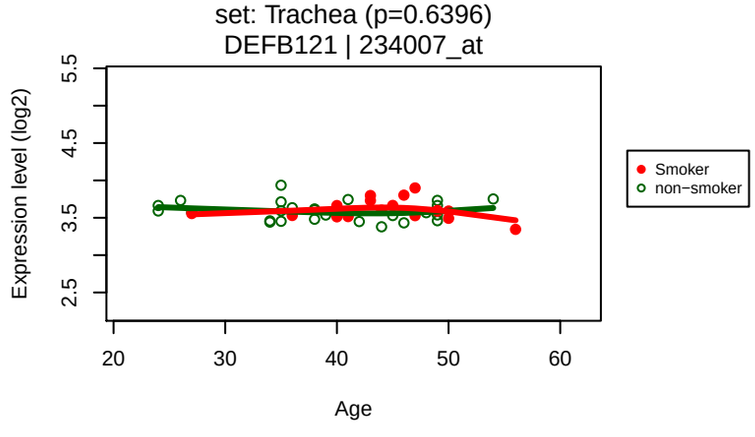

Gene expression ratio in Males
gene: DEFB121 | 234007_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8699
p=0.9309
p=0.4437
p=0.2202
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DEFB121 | 234007_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9045
p=0.7805
p=0.788
p=0.9342
SmallAirway2
LargeAirway2
Trachea
Macrophage2

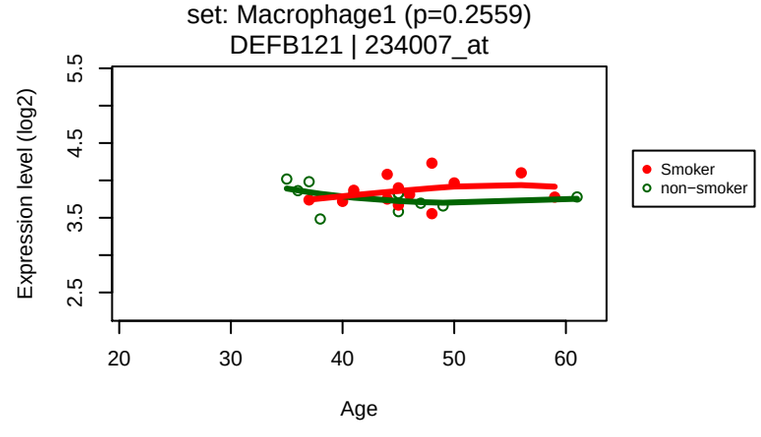

set: SmallAirway1 (p=0.1003)
DEFB121 | 234007_at
Expression level (log2)
Age
Smoker
non−smoker

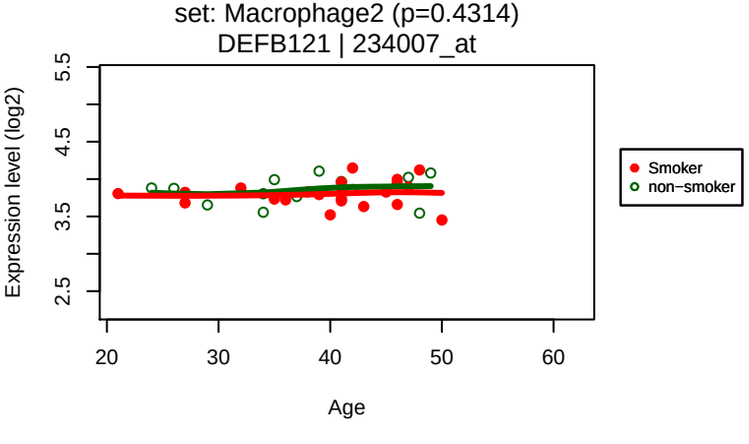

set: SmallAirway2 (p=0.8567)
DEFB121 | 234007_at
Expression level (log2)
Age
Smoker
non−smoker

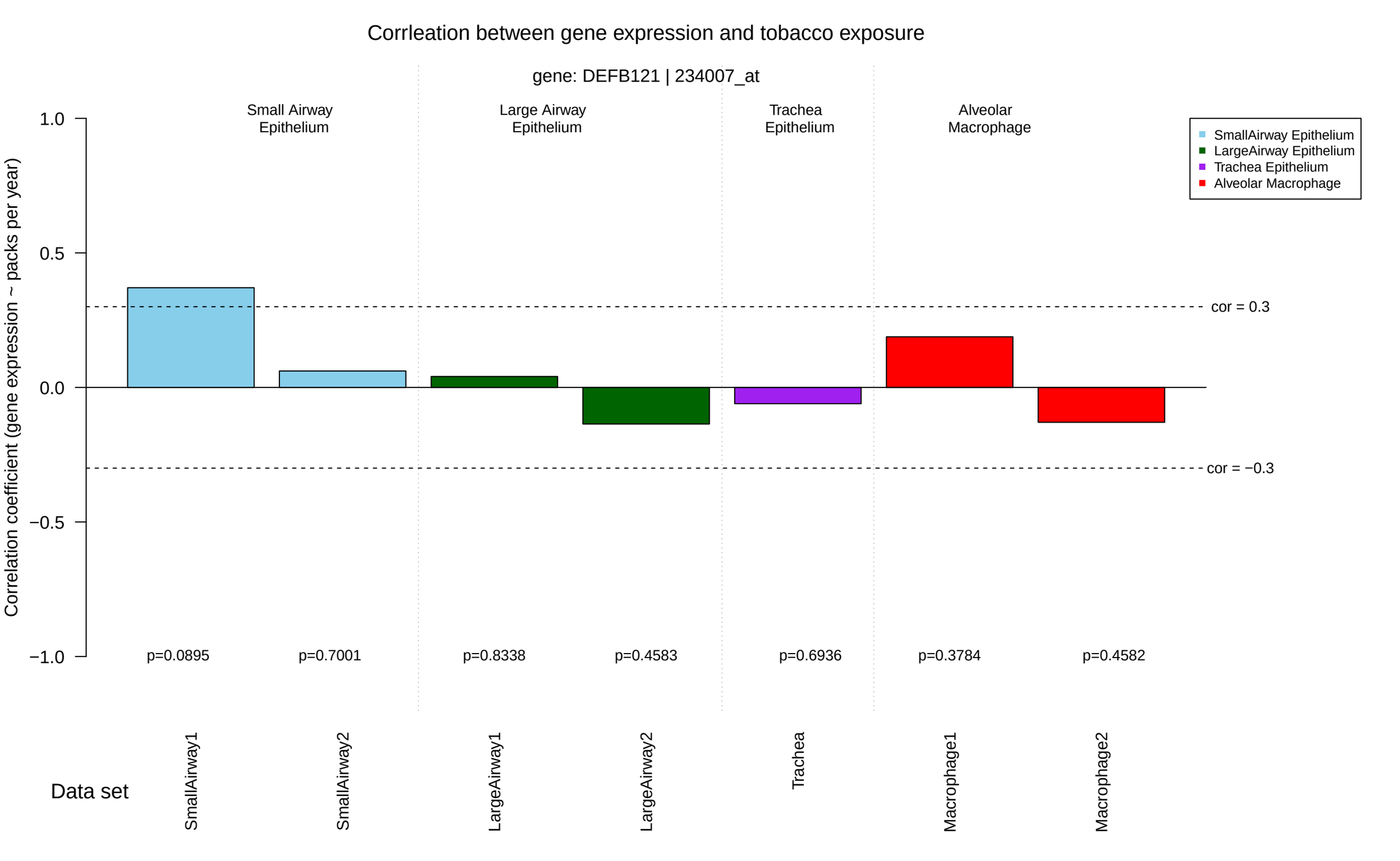

set: LargeAirway1 (p=0.9608)
DEFB121 | 234007_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8202)
DEFB121 | 234007_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6396)
DEFB121 | 234007_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2559)
DEFB121 | 234007_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4314)
DEFB121 | 234007_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DEFB121 | 234007_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0895
p=0.7001
p=0.8338
p=0.4583
p=0.6936
p=0.3784
p=0.4582
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2